

Practice Worksheet: Graphing Square Root Functions

Graph each square root function and describe its characteristics.

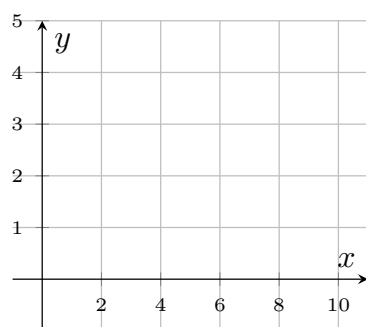
1) $y = \sqrt{x-2} + 1$

Endpoint:

Guide point using a :

y-intercept:

Extra point: (6, ---)



Domain:

Range:

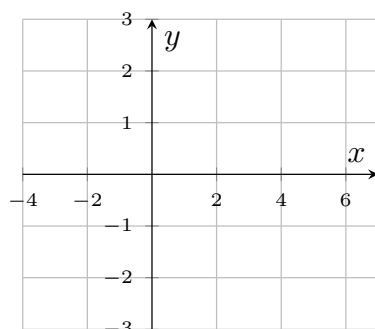
2) $y = \sqrt{x+3} - 2$

Endpoint:

Guide point using a :

y-intercept:

Extra point: (1, ---)



Domain:

Range:

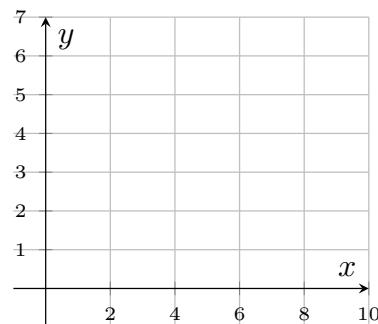
3) $y = 2\sqrt{x}$

Endpoint:

Guide point using a :

y-intercept:

Extra point: (4, ---)



Domain:

Range:

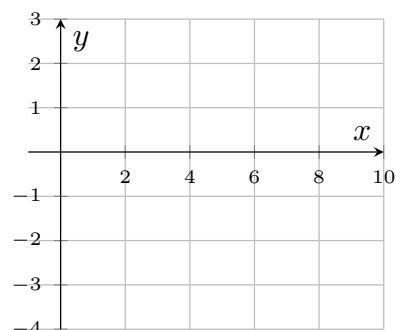
1) 4) $y = -\sqrt{x} + 2$

Endpoint:

Guide point using a :

y-intercept:

Extra point: (4, ---)



Domain:

Range:

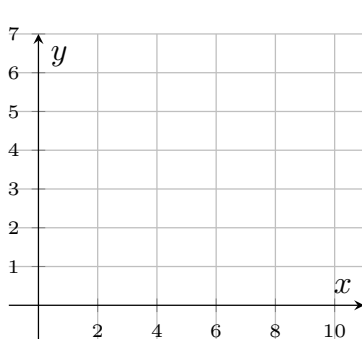
5) $y = \sqrt{x-1} + 3$

Endpoint:

Guide point using a :

y-intercept:

Extra point: (5, ---)



Domain:

Range:

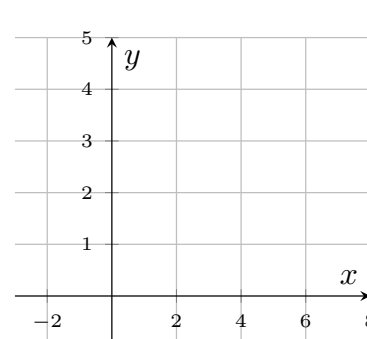
6) $y = -\sqrt{x+2} + 4$

Endpoint:

Guide point using a :

y-intercept:

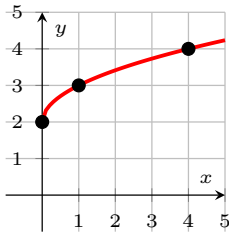
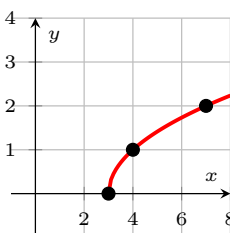
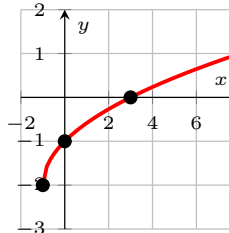
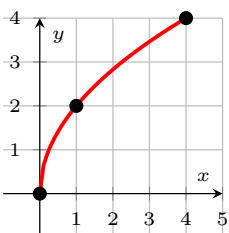
Extra point: (2, ---)



Domain:

Range:

Write the equation of the radical function.

7)  Equation: _____	8)  Equation: _____	9)  Equation: _____	10)  Equation: _____
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Describe each graph as compared to the parent graph $y = \sqrt{x}$.

11) $y = -3\sqrt{x+4} + 5$ The graph of this _____ function has been translated _____ units _____ and translated _____ units to the _____. It has been _____ in the _____-axis and vertically _____ by a factor of _____. It has an endpoint at _____ and is _____ from left to right. The function has a domain of _____ and a range of _____.	12) $y = 2\sqrt{x-1} - 3$ The graph of this _____ function has been translated _____ units _____ and translated _____ unit to the _____. It has been vertically _____ by a factor of _____. It has an endpoint at _____ and is _____ from left to right. The function has a domain of _____ and a range of _____.
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Write the equation that meets the given description. Show all work.

13) A radical function that has an endpoint at the origin and passes through the point (4, 6).	14) A radical function that has a domain of $x \geq -2$ and a range of $y \geq 3$ and was vertically stretched by a factor of 4.	15) A radical function that has been reflected over the x-axis, shifted right 5 units, and shifted down 2 units.
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